

Degradation of Highly Alloyed Metal Halide Perovskite Precursor Inks: Mechanism and Storage Solutions

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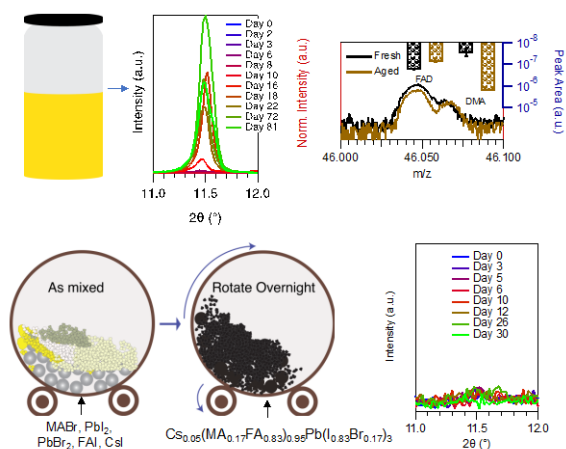
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TOC



Abstract

Whereas the promise of metal halide perovskite (MHP) photovoltaics (PV) is that they can combine high efficiency with solution-processability, the chemistry occurring in precursor inks is largely unexplored. Herein, we investigate the degradation of MHP solutions based on the most widely used solvents, dimethylformamide (DMF) and dimethyl sulfoxide (DMSO). For the MHP inks studied, which contain formamidinium (FA^+), methylammonium (MA^+), cesium (Cs^+), lead (Pb^{2+}), bromide (Br^-) and iodide (I^-), dramatic compositional changes are observed following storage of the inks in nitrogen in the dark. We show hydrolysis of DMF in the precursor solution forms dimethylammonium formate, which subsequently incorporates into the MHP film to compromise the ability of Cs^+ and MA^+ to stabilize FA^+ based MHP. The changes in solution chemistry lead to a modification of the perovskite film stoichiometry, band gap, and structure. The solid precursor salts are stable when ball-milled into a powder, allowing for the storage of large quantities of stoichiometric precursor materials.

Metal halide perovskites (MHP) have the general structure of ABX_3 , where A is a monovalent cation, B is a divalent metal, and X is a halide. There is an extraordinary degree of tunability within this general formula.¹ Alloyed materials with two or even three A-site and X-site constituents have demonstrated advantages over the single cation/halide MHPs such as methylammonium lead triiodide ($MAPbI_3$), formamidinium lead iodide ($FAPbI_3$) or cesium lead triiodide ($CsPbI_3$).^{2,3} Specifically, rational alloys offer the potential to tune the perovskite structure for improved compositional and phase stability,³⁻⁵ enhanced performance⁶ and tunable band gap.^{5,7} Developing scalable, stable precursors for these alloyed materials is important for the industrial production of solution-based MHP optoelectronic technologies. Moreover, the preparation of MHP inks for laboratory-scale research, particularly in the case of complex alloys, requires meticulous measurement of precursor materials to control the fine stoichiometry of the ink. Precise control of stoichiometry is difficult but critically important to reproducible and reliable experimentation. Even slight solution compositional changes could manifest themselves in changes in the batch-to-batch MHP stoichiometry, which modify the resulting materials optoelectronic properties as well as its stability.⁸ As a practical consideration, to prepare a precursor ink for the deposition of highly alloyed MHP such as $Cs_{0.05}(MA_{0.17}FA_{0.83})_{0.95}Pb(I_{0.83}Br_{0.17})_3$ (abbreviated as FAMACs) that Saliba *et al.*³ reported, it is necessary for them to accurately measure 110 mg FAI, 324 mg PbI_2 , 14 mg MABr, 47 mg $PbBr_2$, 11 mg CsI and 512 μ l DMF, 128 μ l DMSO for a typical solution volume required for depositing films in one batch of 16 one-square-inch substrates. As measuring these small amounts of this number of salts and solvents is very tedious and time-consuming, the question of the ink shelf life naturally arises.

In this report, the shelf life of the $Cs_{0.05}(MA_{0.17}FA_{0.83})_{0.95}Pb(I_{0.83}Br_{0.17})_3$ precursor ink reported by Saliba *et al.*³ is explored by fabricating a series of MHP films and devices from a large volume of ink, which was stored in the dark in N_2 . Despite the inert storage conditions, we find that trace water leads to hydrolysis of the DMF, producing dimethylammonium formate in solution. Dimethylammonium cations (DMA^+) are incorporated into the perovskite films and deleteriously affect the optoelectronic properties of the resultant films and devices, which dramatically change the stoichiometry of the films, reduce the amount of MA^+ and Cs^+ , and result in the formation of the yellowish FA^+ - and DMA^+ -based perovskite phases, instead of the black cubic phase of FAMACs. These changes can be partially recovered by adding the deficient cations and halides back into the aged ink, evidenced by the improved optoelectronic performance of the films and devices from the modifications. In the end, mixing the precursor salts in the absence of solvent by ball-milling is demonstrated to effectively avert modification of the perovskite films with precursor storage time.

Degradation of perovskite films from stored solutions

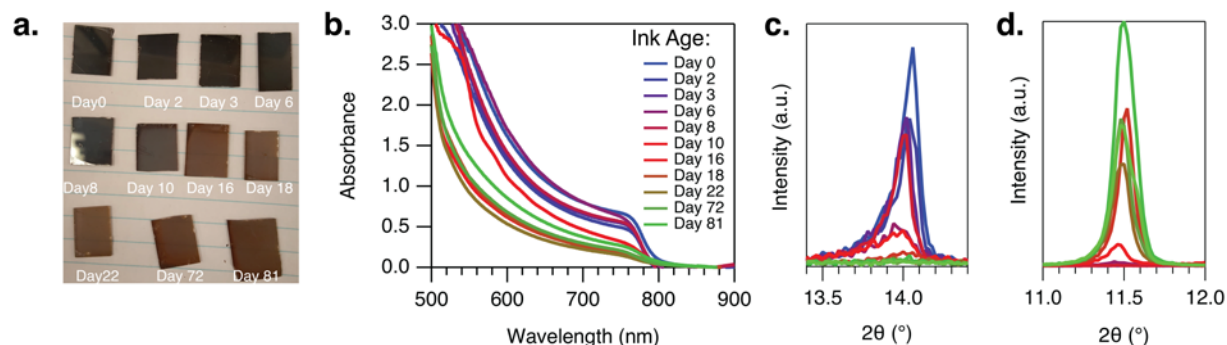


Fig. 1 (a) photographs, (b) absorption spectra, and (c, d) XRD patterns (not normalized) of FAMACs films fabricated from FAMAC ink that was stored for different days in dark in N_2 . Figure (c) and (d) share the same figure legend with (b).

Photographs of FAMACs films fabricated from a fresh ink (*viz.*, 0 days of storage, labeled as “Day 0”), and from the same ink following storage for 2 days to 81 days are shown in Fig. 1a. The appearance of the deposited films clearly becomes visually lighter as the ink gets older, which is confirmed by reduced light absorption (Fig. 1b). Profilometry confirms that this change is not due to film thickness variations (Fig. S1a). In fact, the films get somewhat thicker (average thickness from 542 ± 24 nm on day 0 to 790 ± 84 nm on day 81) with solution aging. Furthermore, the films get rougher (RMS roughness from 7 ± 1.5 nm on day 0 to 16 ± 3 nm on day 81) (Fig. S1b) with ink aging. Because the ink was stored in N_2 and in dark, it is of interest to determine what changed in the films deposited from inks with different ages as well as what changes in the solution resulted in these changes. To understand the structural changes underlying the change in film absorbance, X-ray diffraction (XRD) measurements were obtained (full XRD spectra are presented in Fig. S2). For the FAMACs films that were fabricated with fresh ink, an intense diffraction peak at a 2θ value of 14.0° is observed (Fig. 1c), corresponding to the (100) peak of cubic FAMACs.³ As the ink aged, this peak at 14.0° becomes weaker (Fig. 1c) whereas a peak at 11.5° becomes stronger (Fig. 1d). This latter peak matches to the yellow hexagonal phase of $FAPbX_3$,⁹ but could also be other compounds as will be discussed later.

To understand the effect of these structural changes on solar cell devices (device structure of FTO/ SnO_2 /FAMACs/Spiro-MeOTAD/ $MoOx$ /Al as described elsewhere¹⁰), current density-voltage (J - V) and external quantum efficiency (EQE) measurements were performed (Fig. 2). Statistics of the devices are presented in Fig. S3. The devices made with fresh ink (“Day 0”) show 18.2% power conversion efficiency (PCE) with a short-circuit current (J_{sc}) of 21.13 mA cm^{-2} , an open-circuit voltage (V_{oc}) of 1.09 V and a fill factor (FF) of 77.6%, when measured using a reverse scan (full J - V scans are presented in Fig.

S3). Devices from ink aged for 1 day do not show significant change from devices with fresh ink, but devices from a 5-day-age ink show decrease in J_{SC} (19.40 mA cm⁻²) and FF (73.6%) along with an improved V_{OC} (1.13 V). The ~10 nm blue shift in the EQE onset for this device suggests that the decreased J_{SC} and increased V_{OC} , at least in part, due to active layer composition change such as having less I⁻ incorporation¹¹ or smaller A-site cations in the lead halide perovskites.^{11,12} After aging the ink for 52 days the device performance has dropped to 4.28% PCE, with all photovoltaic parameters (J_{SC} = 10.22 mA cm⁻², V_{OC} = 0.95 V, FF = 43.2%) decreased.

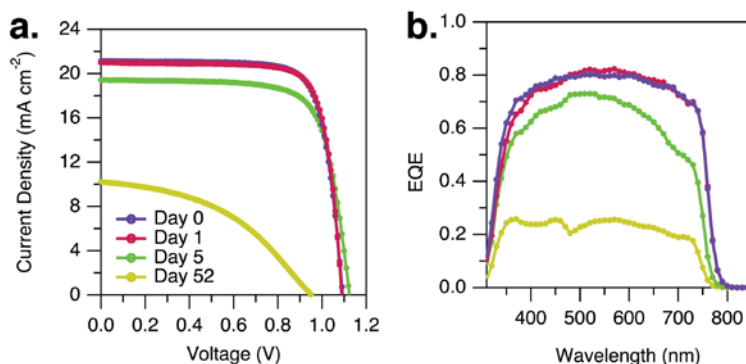


Fig. 2 (a) Open circuit voltage to short-circuit current scanned (reverse scan) J - V curves and (b) EQE for the best devices fabricated with FAMACs ink stored for different times: fresh ink, 1 day, 5 days and 52 days.

Proposed mechanism of precursor solution degradation

Fourier transform infrared (FTIR) spectroscopy is used to study the organic constituents of the FAMACs films to obtain mechanistic insight into the precursor solution degradation. Fig. 3 shows FTIR spectra for two films that are respectively made from a fresh ink and an ink with 240 days aging (stored in dark, N₂ environment). FTIR spectra are recorded before annealing and after annealing of the films. Data in Fig. 3a, clearly shows the presence of FA in all films with the prominent FTIR peak at 1712 cm⁻¹ due to strong C=N symmetric stretching. In addition to this characteristic peak, there are peaks at 3407 cm⁻¹, 3359 cm⁻¹, 3272 cm⁻¹ and 3171 cm⁻¹ which are due to symmetric N-H stretching in the ammonium cations of the lattice¹³. However, in addition to these peaks, the films cast from the aged ink exhibit FTIR peaks (Fig. 3b) at 1430 cm⁻¹ and 1380 cm⁻¹, which can be attributed to the characteristic rocking and symmetric vibrational modes, respectively, of the formate anion (HCOO⁻).¹⁴

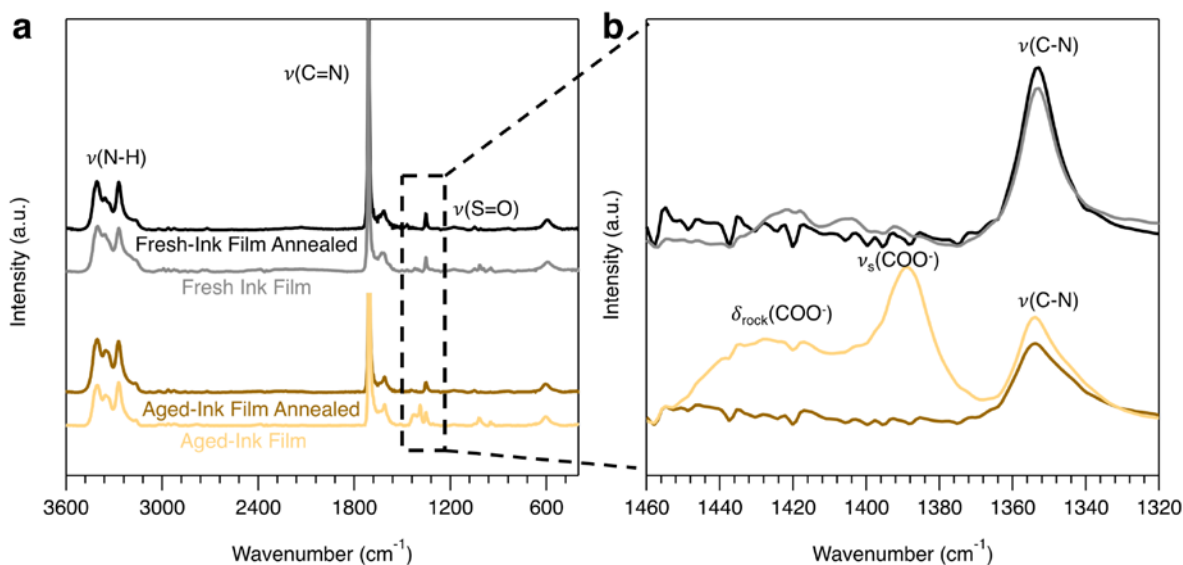
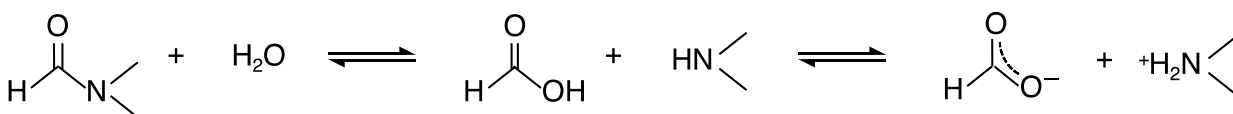


Fig. 3 (a) FTIR spectra of FAMACs films that are fabricated from fresh ink (black lines) and 240 Days old ink (labeled as “Aged-Ink Films”, yellow lines). Spectra offset for clarity. (b) COO⁻ and C-N vibrations. Lighter lines are before annealing and darker lines are after annealing. All spectra baseline corrected and normalized to the intensity of the ν(C=N) vibration.

Recent work has shown that DMF will readily hydrolyze at room temperature under acidic conditions when coordinated with a metal center.¹⁵ DMF coordination to Pb²⁺ compounds in perovskite precursors is well known to occur.¹⁶ The hydrolysis of DMF yields formic acid (HCOOH) and dimethylamine (DMA), as shown in Scheme 1. Interestingly, formic acid was recently shown to have important ramifications for the optoelectronic quality of metal halide perovskite films¹⁷ and DMA has shown to be able to incorporated into lead halide perovskite structure^{18–20}. DMA has also been shown to form in perovskite solutions due to the acid-catalyzed reaction between DMF and methylamine to yield DMA and N-methylformamide.²¹ DMA is a stronger Lewis base than DMF or DMSO and will form stable complexes with Lewis acids such as Cs⁺, Pb²⁺, and PbX⁺.²² The equilibrium of the precursor solution complexes will thus be affected by DMA and formic acid, DMA⁺ and formate (HCOO⁻). HCOO⁻ will interact in stronger fashion than halides as a bidentate ligand with both Cs⁺ and Pb²⁺. HCOO⁻ clearly remains visible by FTIR of the aged films before annealing, despite the antisolvent treatment during film deposition. Within the sensitivity of the FTIR, HCOO⁻ appears to be removed with annealing.



Scheme 1. Proposed ink aging mechanism. Hydrolysis of DMF to formic acid and dimethylamine (DMA) and protonation of DMA for produce formate (HCOO^-) and dimethylammonium (DMA^+).

To further investigate the proposed mechanism, Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS) was used to probe the composition of perovskite films cast from a fresh and a 30-day old ink (Fig. 4a-i). This technique provides information on all of the chemical species in the films, including the organic components. The ToF-SIMS profiles for MA^+ , FA^+ , FAD (D=deuterium), DMA^+ , Br^- , DMABr , I^- , DMAI , $^{204}\text{Pb}^{2+}$ and Cs_2^+ which correspond, respectively, to m/z ratios of 32.050, 45.046, 46.046, 46.066, 78.911, 124.890, 126.893, 172.890, 203.959, 265.799 were analyzed (Table 1). The obtained ToF-SIMS traces were normalized to the total ion beam counts and integrated through the entire film thickness to obtain a total signal intensity at each m/z . The components in the two samples can then be compared by the integrated peak area at the m/z ratio of interest. At each m/z of interest, the ToF-SIMS signal peak area is fitted, through a combination of Gaussian and Lorentzian convolution²³ with linear background, and presented as bars in Fig. 4(a-g). Fitting results and its standard deviations are shown with the peak area bars.

Table 1. Assignment of key peaks in the ToF-SIMS spectra

m/z	Fragment ^α	Ionic Charge ^β
32.050	MA	MA^+
45.046	FA	FA^+
46.046	FAD	FAD^+
46.066	DMA	DMA^+
78.911	Br	Br^-
124.890	DMABr	DMABr
126.893	I	I^-
172.890	DMAI	DMAI
203.959	Pb	Pb^{2+}
265.799	Cs	Cs^+

^α “Fragment” corresponds to the component tracked by mass spectrometry (all are positive ions)

^β “Ionic Charge” corresponds to the ionic charge of the component in the perovskite film

Compared to the fresh-ink film, a slight decrease in the ToF-SIMS signal intensity is observed in the 30 days-aged-ink films in all cations, including MA^+ , FA^+ , Cs^+ and Pb^{2+} , whereas the changes in the I^- and Br^- signals are not statistically significant. Importantly, in addition to these expected signals, the aged sample shows an additional peak at m/z ratios of 46.066 (Figure 4c). We attribute this additional peak to DMA^+ , which appear in the mass spectra as a result of DMF hydrolysis. Stronger ToF-SIMS signals are observed at m/z of 124.89 and 172.89, which are attributed to DMABr and DMAI (Table 1) respectively, further confirming the increased content of DMA^+ in aged ink film.

Using X-site anions' ($\text{X} = \text{I}^- + \text{Br}^-$) ToF-SIMS signal intensity, which did not change much with aging, ratios of MA/X , FA/X , DMA/X , Br/X , DMABr/X , I/X , DMAI/X , Pb/X , Cs/X are computed and presented in Fig. 4h. These ratios confirm the proposed scheme where DMA^+ has been incorporated into the perovskite film, which is consistent with the increasing film thickness with ink aging. DMA^+ incorporation harms the optoelectronic properties of the resulting films and the photovoltaic performance of devices.

The ink aging data shown in Fig. 1 and Fig. 2, indicates a blue shift (~ 10 nm) in absorption onset and EQE in the resulting films. One possible reason for this shift could be proportionally less I^- incorporation; however, as no significant X-site content (I^- and Br^-) changes were observed in the TOF-SIMS, it is unlikely the absorption blue shift is the result of X-site changes. Recently, multiple groups^{18–20} have shown a wide optical bandgap for dimethylammonium lead halide (DMAPbX_3): 2.39 eV for DMAPbI_3 and 3.03 eV for DMAPbBr_3 by Malasavi *et al.*¹⁹, 2.59 eV for DMAPbI_3 by Senaris-Rodriguez *et al.*¹⁸, and 2.58 eV for DMAPbI_3 by Tao *et al.*²⁰ DMA^+ incorporation into the deposited MHP films would explain the absorption onset and EQE blue shift that is observed in Fig. 1 and Fig. 2. Structurally, the characteristic XRD peak for DMAPbX_3 lies at 2θ of 11.6° , which is very close to the characteristic XRD peak of the yellow, hexagonal phase of FAPbX_3 . Therefore, the increasing XRD peak at $\sim 11.5^\circ$ with ink aging (Fig. 1d) could be due to either hexagonal FAPbX_3 and/or DMAPbX_3 .

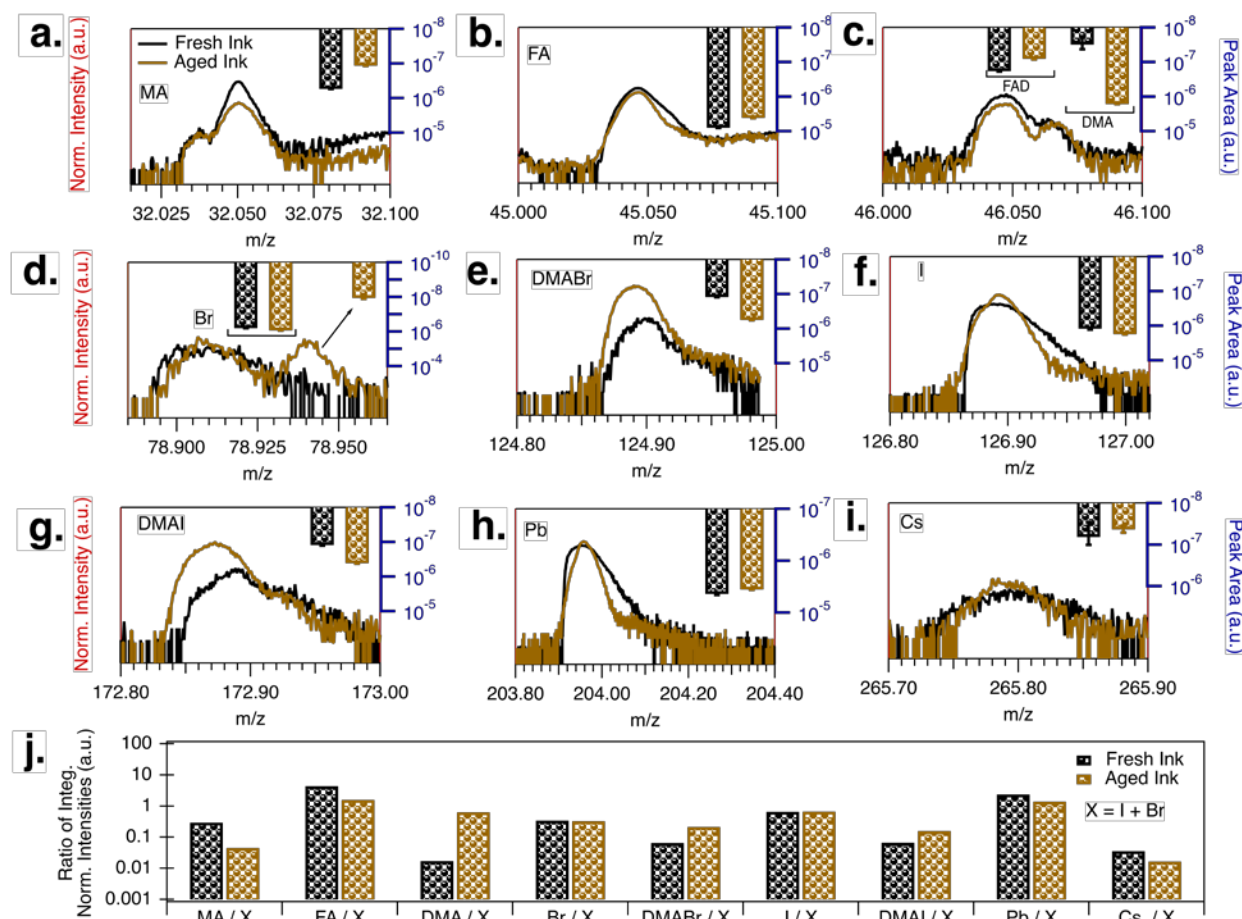


Fig. 4 ToF-SIMS spectra of FAMACs films from fresh ink (black lines and bars) and 30 days aged ink (labeled as “Aged Ink”, brown lines and bars) from normalized ToF-SIMS signals: (a) MA; (b) FA; (c) DMA; (d) Br; (e) DMABr; (f) I; (g) DMAI; (h) Pb; (i) Cs. Bars in each figure shows the integrated peak area, fitted through a combination of Gaussian and Lorentzian²³, of ToF-SIMS signal for each elements. The error bars are standard deviations of the fits. (j) Ratios of MA/ X, FA/ X, DMA/X, Br/X, DMABr/X, I/X, DMAI/X, Pb/ X, Cs/ X from integrated peak area of ToF-SIMS.

Recovery of the aged ink by adding various instable cation and halides

We confirmed that the preferential formation of the hexagonal phase of FAPbX_3 with DMA^+ is related to the reduced incorporation of MA^+ and Cs^+ , as evidenced by ToF-SIMS, as well as the larger presence of DMA^+ . To recover an aged ink and reduce the amount of DMA^+ in the film, one strategy we applied was to add MA^+ - and Cs^+ - containing salts into the aged ink (24 days old) to compensate the loss of MA^+ , Cs^+ in FA^+ based perovskite. The added perovskite salts include: (1) MABr and CsI , as they are the primary source of MA^+ and Cs^+ , (2) pure MABr , (3) pure CsI to decouple the effect of MA^+ and Cs^+ ,

and (4) pure MAI to have excess I^- to address bandgap shift²⁴ induced by the DMA⁺ into FA⁺-based framework. For simplicity, the quantity of each perovskite salt added to the aged ink is equal to the full amount of each perovskite salt (with Cs⁺ and MA⁺ source) used to formulate the fresh ink. The specific amount of the salts added is detailed in Table S1 and Table S2. FAMACs films were fabricated using these modified inks. Various characterization data is shown in Fig. 5. From the photograph of films shown in Fig. 5a, it is obvious that the films become darker in appearance for all modifications. The UV-vis absorption spectra (Fig. 5b) show that no absorption edge shift, compared to unmodified aged ink based films, is discernable in the films from modified inks with MABr and CsI, pure MABr, or pure CsI. Conversely, an absorption edge red shift, towards the absorption edge of films deposited from fresh ink, is observed for films cast from the 24 days old ink with additional MAI added. A similar red shift is observed by photoluminescence (PL) spectrum shown in Fig. 5c, likely indicating the excess I^- in the solution has effectively changed the bandgap of the resulting film.

The effect of aged ink modification was further investigated by examining structural changes with XRD. Consistent with absorption data, all of these ink modifications suppress the XRD peak at 11.5° and increase the peak intensity at 14.0°; however, none of the modifications made were able to fully suppress the 11.5° peak (Fig. 5d, Fig. 5e). Having found that adding MA⁺ and Cs⁺ containing perovskite salts into aged ink will increase the absorbance of films by suppressing the yellowish DMA⁺ incorporated FAPbX₃, it can be expected that solar cells fabricated from aged FAMACs ink with additional salts will perform better than the devices from the same ink without modification. Fig. 5f shows the device performance for the various ink modifications attempted. In all cases, the device performance did improve, PCE improved from 5.68% to 9.86% with MABr and CsI addition; to 8.20% with MABr addition; to 10.03% with CsI addition; or to 9.86% with MAI respectively. The improvement is primarily the result of increased J_{sc}. While the devices did improve with these modifications, a recovery approaching the performance of the fresh samples did not occur, consistent with the formation and incorporation of DMA⁺ into the films which cannot be fully remedied by changes to the precursor ink stoichiometry.

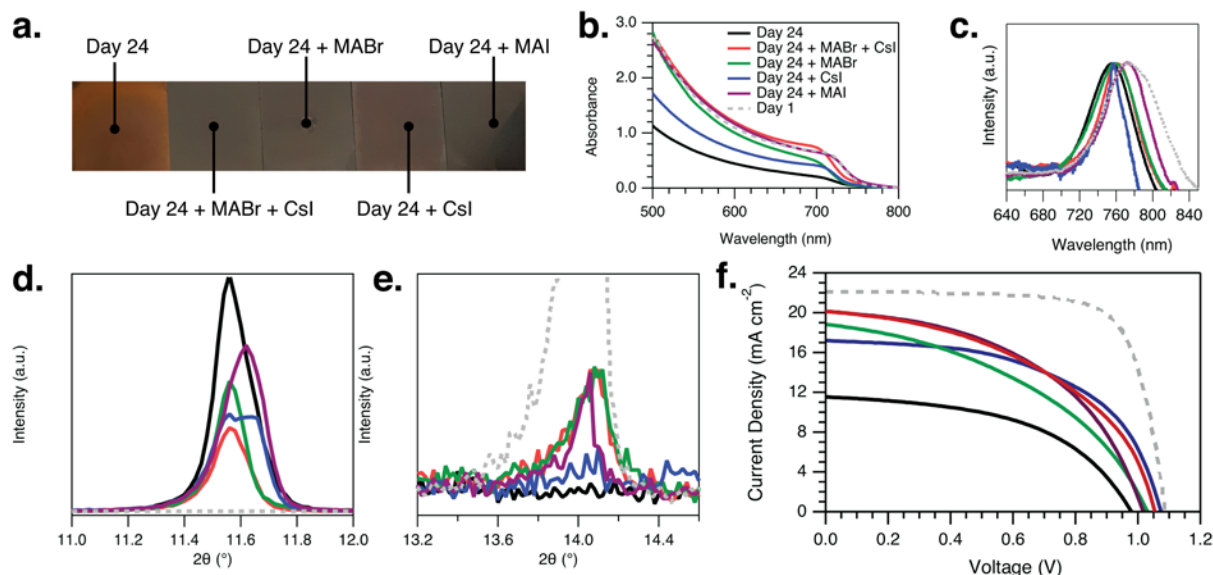


Fig. 5 Characterizations of films that were fabricated from 24 days old ink (black solid lines), and 24 old ink with newly added MABr and CsI (red solid lines), or MABr (green solid lines), or CsI (blues solid lines), or MAI (purple solid lines): (a) photographs, (b) UV-vis absorption spectrum, (c) PL spectrum, (d) (e) XRD patterns (not normalized) and (f) J-V curves. The film made from fresh ink (grey dash lines, 1-3 hours between ink preparation and film deposition) is included as a reference.

Storage for highly alloyed Ball-Milled MHP powders

It is clear that long-term storage of DMF-based FAMACs perovskite inks is challenging, yet the ability to store these inks remains highly desirable. Naturally, the question arises: what is a practical way to store these precursor salts to avoid inconsistent ink preparation which adversely affect results and reduce reproducibility? In light of our understanding of the ink degradation mechanism, we propose the storage of the salts in the absence of solvents which may degrade. Solvent-less storage can be achieved by utilizing a ball mill to uniformly mix the various salts into a single powder of the desired stoichiometry, followed by storage as a solid powder. The solid powder can then be dissolved in DMF and DMSO immediately before use. Critically, it should be noted in this experiment, the DMF and DMSO solvent are still aging. To minimize this, the solvents are stored in N_2 .

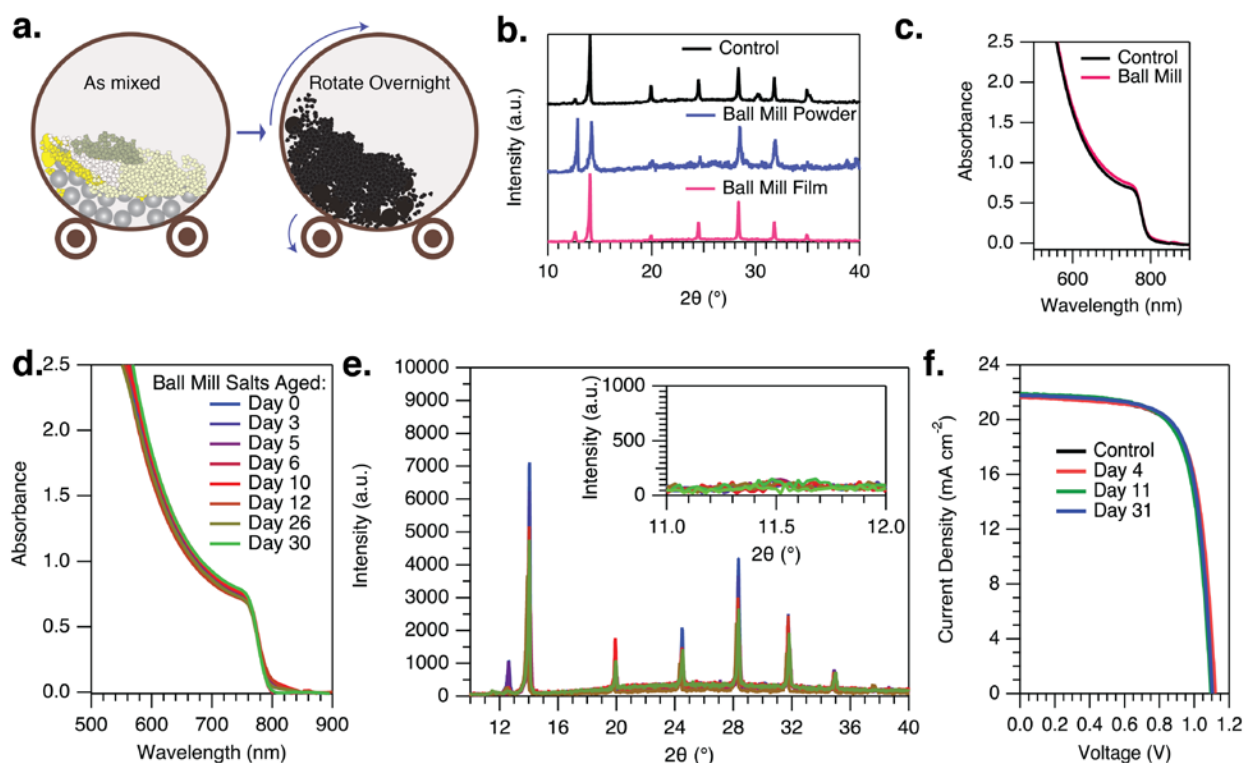


Fig. 6 (a) Illustration of ball-mill process; (b) XRD of Day 0 FAMACs film (control), ball-milled powder and fresh film from ball-milled precursor; (c) absorptions of day 0 FAMACs film and fresh film from ball-milled precursors; (d) Absorption spectra and (e) XRD patterns of FAMACs films made from ball-milled salts that were stored for different times (up to 31 days). (f) *J-V* characterization of champion FAMACs devices with FAMACs films made from ball-milled salts that were stored for different times.

Fig. 6a shows the schematic of the ball mill setup. The precursor salts are mixed in the desired stoichiometry and ball-milled overnight to form a black powder due to formation of the perovskite, as confirmed by XRD (Fig. 6b). The strong XRD peak at 12.5° in ball-milled powder is ascribed to the (001) peak of unreacted PbI_2 . The mixed perovskite powders can be readily dissolved in DMF and DMSO at the desired concentration before fabricating FAMACs films. Perovskite films prepared from this ball-milled powder show no discernable differences in either structure or absorption (Fig. 6c) when compared to the standard ink preparation procedure. To confirm that these mixed salts are stable, they were stored for up to 31 days with absorption (Fig. 6d) and XRD patterns (Fig. 6e) recorded on films fabricated intermittently throughout the process. Within the timeframe of the study, no degradation in either the salt mixture or the resultant film quality is observed. Importantly, the XRD peak at 11.5° is noticeably absent, i.e. hexagonal phase of FAPbX_3 or DMAPbI_3 (see Fig. 6e inset). Furthermore, FAMACs perovskite devices (with the structure, $\text{FTO}/\text{SnO}_2/\text{FAMACs}/\text{Spiro-MeOTAD}/\text{MoO}_x/\text{Al}$) fabricated throughout the

course of this experiment show no statistically significant changes in performance relative to each other or control devices (Fig. 6f and Fig. S4).

Conclusion

In summary, the chemistry occurring in standard perovskite inks over time is elucidated and shown to modulate the composition and crystal phase of the resulting FAMACs perovskite films. With ink aging, we find that DMA⁺ incorporates into the perovskite film, which, along with a reduction of MA⁺ and Cs⁺ cations, results in a film that appears yellowish and performs very poorly as a photovoltaic absorber. DMA⁺ is formed as a result of the hydrolysis of DMF by trace water, catalyzed by an acid catalyst, most likely Cs⁺, Pb²⁺, or PbX⁺. The ink aging-induced stoichiometry in the MHP film can be offset to a moderate degree by adding MA⁺-, Cs⁺-, and I⁻-containing salts, evidenced by improved optoelectronic properties of resulting films and devices. To address ink aging-induced phase instability in FAMACs perovskites, a method of storing ball-milled salts is proposed and validated with various optoelectronic characterizations of FAMACs films and devices from ball mill salts aged for over 30 days. This study highlighted the degradation in highly alloyed perovskites inks relevant to their use in commercialization of MHP-based optoelectronics and offered an effective alternative method of precursor storage, which can readily be implemented at lab or industrial scale.

Supporting Information

Experimental methods, thickness and roughness measurement, XRD, device performance statistics.

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